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Oligosaccharides

Rational Design and Expedient Synthesis of Heparan Sulfate Mimetics from Natural Aminoglycosides for Structure and Activity Relationship Studies

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Abstract: Heparan sulfate (HS) contains variably repeating disaccharide units organized into high- and low-sulfated domains. This rich structural diversity enables HS to interact with many proteins and regulate key signaling pathways. Efforts to understand structure-function relationships and harness the therapeutic potential of HS are hindered by the inability to synthesize an extensive library of well-defined HS structures. We herein report a rational and expedient approach to access a library of 27 oligosaccharides from natural aminoglycosides as HS mimetics in 7–12 steps. This strategy significantly reduces the number of steps as compared to the traditional synthesis of HS oligosaccharides from monosaccharide building blocks. Combined with computational insight, we identify a new class of four trisaccharide compounds derived from the aminoglycoside tobramycin that mimic natural HS and have a strong binding to heparanase but a low affinity for off-target platelet factor-4 protein.

Introduction

Heparan sulfate (HS) is one of the most abundant glycosaminoglycans (GAGs), which are linear, negatively charged polysaccharides of high molecular weight up to 100000 Dalton (Figure 1).[1] Heparan sulfate is found at the cell surface and extracellular matrix (ECM).[2] It is composed of repeating disaccharide units of D-glucosamine (GlcN) and uronic acid, where uronic acid can be either D-glucuronic acid (GlcA) or L-iduronic acid (IdoA) (Figure 1).[3] The GlcN unit is connected to the uronic acid unit (GlcA or IdoA) through an α(1,4)-glycosidic linkage, while GlcA/IdoA is connected to GlcN through a β(1,4)-linkage (Figure 1). Heparan sulfate is structurally related to the anticoagulant heparin. While HS is located outside the cell, heparin is inside mast cells.[3] The most common disaccharide unit within HS comprises a GlcA linked to GlcN, making up 50 % of the total disaccharide units.[4] In contrast, GlcN-IdoA disaccharide units make up 75 % of heparin.

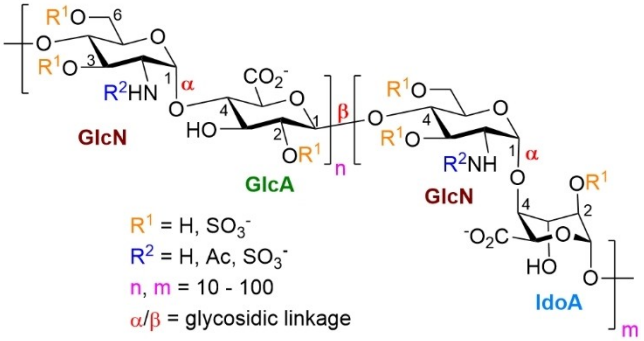


Figure 1. Structure of heparan sulfate polysaccharide.

Sulfation is a dynamic and complex posttranslational modification process. It can occur at the *N*-, 6-*O*-, and 3-*O*-positions of GlcN and the 2-*O*-position of GlcA or IdoA of HS (Figure 1), giving rise to variably sulfated patterns. As a consequence of its polyanionic character, HS interacts with hundreds of proteins and plays a crucial role in many biological processes.[5] Changes in the degree of sulfation (either under or over-sulfation) are also associated with several diseases[6] and may direct the location or activities of proteins.[7] Efforts to decipher structure-function relationships of HS-protein interactions and develop the therapeutic potential of HS-mimicking oligosaccharides[8] and glycopolymers[9] are hindered by the complexity and micro-heterogeneity of HS and a lack of tools to manipulate specific HS-protein interactions.

To advance our understanding of HS biology, it is necessary to access structurally diverse HS oligosaccharides with defined sulfation domains in large quantities. Chemical synthesis has been successful in providing solutions to this challenge.[10] Several HS molecules with specific sulfation sequences have been synthesized to resemble the antithrombin III (ATIII)-binding sequence,[11] to be developed into an antithrombotic drug,[12] to find use as a therapeutic agent for Alzheimer's disease,[13] to inhibit the herpes simplex virus type 1 host-cell interaction,[14] and to modulate heparanase activity.[15] Despite this remarkable progress, accessing a diverse collection of HS-mimicking molecules lies in preparing selectively protected HS monosaccharides, which are not commercially available and were synthesized in 7–11 steps (Figure 2A).[12–15] As such, even syntheses of selectively

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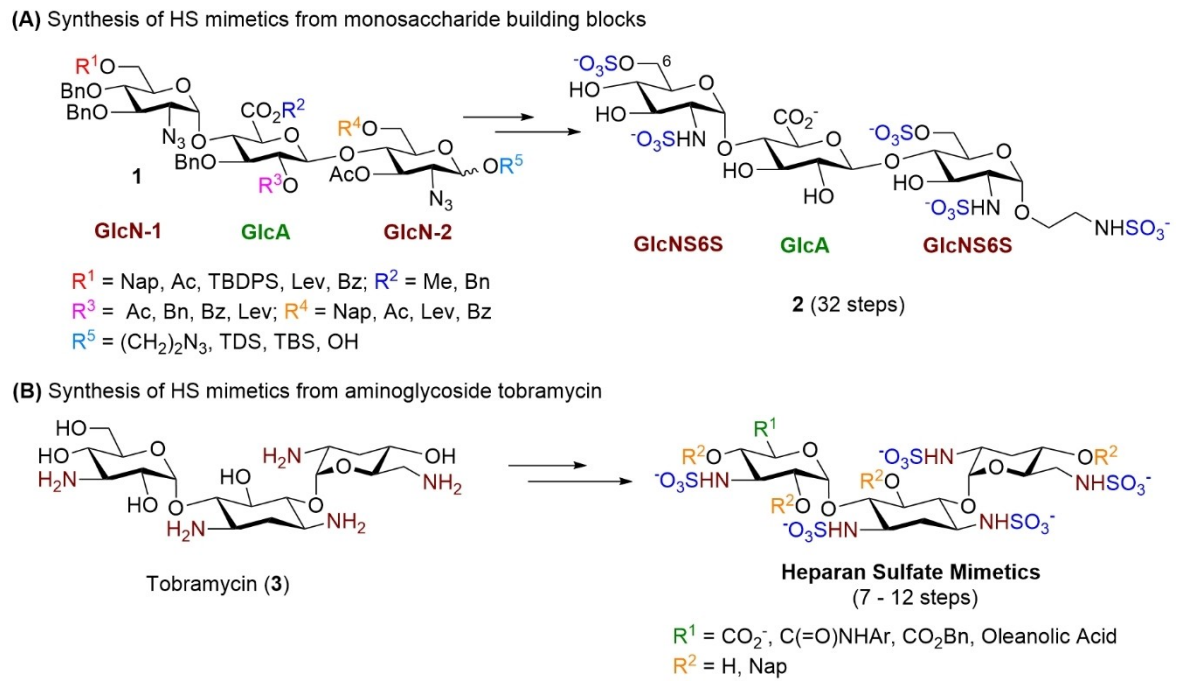


Figure 2. Approach to HS oligosaccharides and HS-mimicking oligosaccharides. A) Synthesis of selectively protected trisaccharide intermediates and HS trisaccharide **2** from traditional monosaccharide precursors. B) Synthesis of oligosaccharides resembling HS from the commercially available aminoglycoside tobramycin **3** with Na^+ as a counterion.

protected GlcN-GlcA-GlcN trisaccharide intermediates require 24–34 steps.[12–16] Recently, the core disaccharides of HS were elegantly generated from natural heparin, *N*-Ac-heparin, and heparosan.[17,18] While this strategy is indisputably a powerful tool to expedite HS synthesis, it requires an estimated 10–12 steps to convert a fully acetylated GlcN-IdoA into selectively protected GlcN-GlcA if GlcNS6S-GlcA disaccharide is the required core (e.g., **2** in Figure 2A) for the assembly of HS oligosaccharides.[17]

Herein, we report an expedient strategy to rationally design and synthesize an extensive library of 27 sulfated trisaccharides as HS mimetics from low-cost and readily available aminoglycoside tobramycin (**3**, Figure 2B). This strategy significantly reduces the total number of steps by divergent modification of a single protected trisaccharide intermediate derived in 3 steps from tobramycin. Syntheses of trisaccharides resembling HS, with defined sulfation domains and varied functional groups, are achieved in 7–12 steps from tobramycin (Figure 2B). These HS mimetics were studied for their ability to interact with heparanase and platelet factor 4 (PF4) protein, which causes antibody-induced thrombocytopenia associated with life-threatening thrombosis.[19,20] Mammalian heparanase is upregulated in the tumor microenvironment of all types of cancers.[21] It degrades polymeric HS into small oligosaccharides bound to proangiogenic and protumorigenic growth factors. Thus, heparanase has been viewed as a promising target for anti-cancer drug development.[22] However, anti-heparanase-based therapy has not yet been implemented in the clinic. All anti-heparanase compounds being examined in clinical trials are heparin-like compounds. While these molecules have been successful in many in vivo models, bleeding and thrombotic complications halted their clinical advancement in humans.[23] Upon evaluation of 27 HS mimetics, four compounds exhibited strong binding to heparanase and had a weak affinity for off-target PF4 protein due to their *N*-sulfation and carefully positioned hydrophobic side chains on the tobramycin saccharide framework. They are more potent heparanase inhibitors than trisaccharide **2** (Figure 2A), prepared in 32 steps from non-commercially available monosaccharide building blocks.[15] This expedient strategy highlights opportunities for the rapid generation of HS mimetics to decipher HS-protein interactions and develop HS therapeutic potential.

Results and Discussion

Our goal is to develop cost-effective strategies, aided by computational insight, for the expedient and scalable preparation of HS-mimicking trisaccharides with a high affinity for heparanase and a low affinity for off-target PF4 protein.[19] Aminoglycosides are attractive in this light as they are commercially available (Figure 3).[24] They target 16S bacterial rRNA and inhibit protein synthesis.[24] They contain multiple hydroxyl and amine groups and have a high affinity toward negatively charged nucleotides at physiological pH 7.4.[24] We hypothesize that if the hydroxyl and amine groups of aminoglycosides are selectively sulfated, the resulting sulfated aminoglycans will no longer bind to rRNA but can interact with heparanase. With the aid of computational study, we identified three *N*-sulfated amino-

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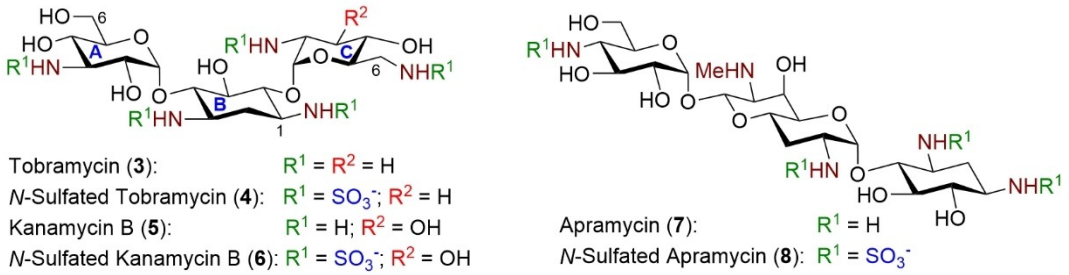


Figure 3. Tobramycin (**3**), kanamycin B (**5**), and apramycin (**7**) and their *N*-sulfated aminoglycans **4**, **6**, and **8** with Na^+ as a counterion.

glycosides derived from tobramycin (**3**), kanamycin B (**5**), and apramycin (**7**; Figures 3 and S4).

To validate our design, we initiated in-silico docking studies with *N*-sulfated aminoglycosides into the *apo* heparanase crystal structure (PDB code: 5E8M)[25] imported into the AutoDock Vina Plugin of the YASARA program (Figures S3, S4).[26] We used HS trisaccharide **2** (Figure 2A) as a model substrate to determine its location in the binding pocket and utilize it as a benchmark for comparison to designed compounds. Compound **2** was placed into the binding pocket of heparanase in the same direction as natural HS trisaccharide (Figure S8).[25] The ligand-heparanase complex was subjected to molecular dynamics (MD) simulations to validate the structure's stability. The final conformation adopted by **2** has strong hydrophobic interactions along the sugar ring, consequently pushing it towards Gly350 and Gly349 (Figure 4A). Towards the end of the MD simulation, **2** lost the interactions with the inner pocket residues, and its upper part (located near Gly350) moved out from the inner pocket, opening more to the solvent environment (Figure 4A). In addition, **2** does not have a consistent set of high occupancy hydrogen bonds (Table S6). Nevertheless, the GlcA CO_2^- group of **2** forms a strong hydrogen bond with Lys232, while the 6-*O*-SO_3^- of the GlcN units was involved in hydrogen bonds with Arg272, Gln270, and Arg303 and salt bridges with Arg272 and Arg303 (Figure 4A). This set of interactions provides a baseline of key interactions required to achieve the binding potency of **2**.

Next, the binding of *N*-sulfated tobramycin **4** into the *apo* crystal structure of heparanase was explored computationally. Computational analysis showed that **4** (Figure 4B) is oriented in the same direction as HS trisaccharide **2** (Figure 4A) in the binding pocket of heparanase. However, **4** had a lower overall hydrogen bonding occupancy (Table S4), especially toward the end of the trajectory, compared to **2**. In addition, compound **4** lost some interaction with Glu343 located deep inside the binding pocket, compensated for by a new strong interaction with Glu225. However, **4** maintained hydrogen bond and salt bridge interactions with Arg272 and Lys232.

To validate our computational results and determine if structural differences between **2** and sulfated aminoglycosides would correspond to in vitro changes in heparanase inhibition, *N*-sulfated tobramycin **4**, kanamycin **6**, and apramycin **8** were prepared (Schemes S1–S4 in the Supporting Information).[27] The heparanase-inhibiting activity of **4**, **6**, and **8** was then evaluated using the in vitro TR-FRET

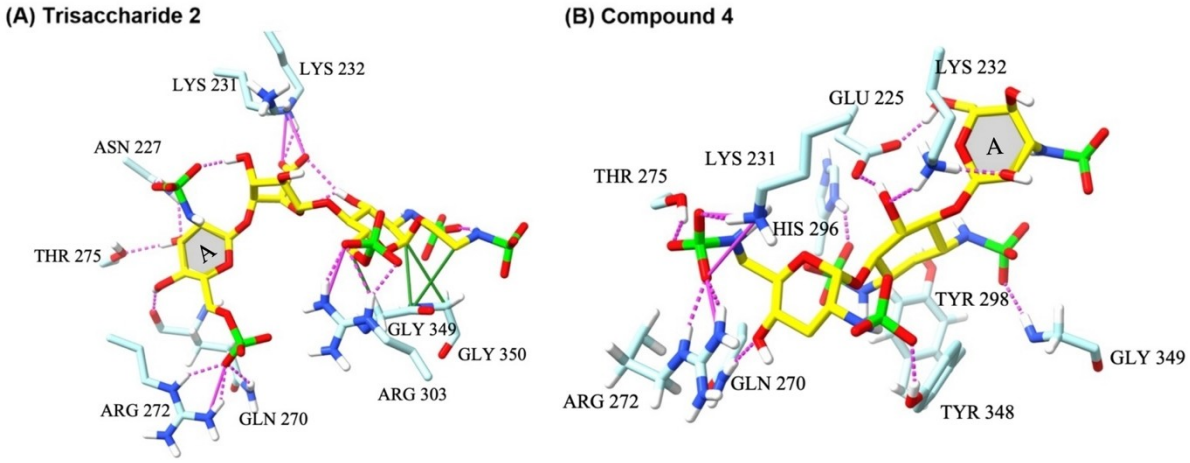


Figure 4. Computational docking of HS mimetics. A) Key interactions between HS trisaccharide **2** for a representative snapshot at the end of the MD simulation and active site residues. B) Interactions between HS mimetic **4** for a representative snapshot at the end of the MD simulation and active site residues. *Color code description for heparanase-ligand interactions*: dashed magenta (hydrogen bond), solid magenta (salt bridge), solid green (hydrophobic interaction), solid blue (pi-cation), yellow stick (HS mimetics), light blue stick (active site amino acid residues). For simplicity, we only labeled ring **A** of the saccharide moiety.

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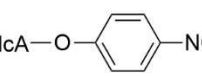
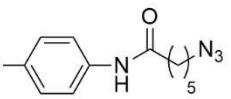
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assay, which is based on the time-resolved (TR) measurement of fluorescence and fluorescence resonance energy transfer (FRET).[28] As shown in Table 1, *N*-sulfated tobramycin **4** was the most potent compound against heparanase with an IC_{50} value of 0.88 ± 0.030 μM (entry 2) but less potent than HS trisaccharide **2** ($IC_{50} = 0.39 \pm 0.010$ μM, entry 1). This result is consistent with our computational studies (Figure 4). However, it is essential to note that it took only 6 steps to prepare **4** and 32 steps to prepare **2**.[15] Both *N*-sulfated kanamycin **6** ($IC_{50} = 5.9 \pm 0.77$ μM, entry 3) and *N*-sulfated apramycin **8** ($IC_{50} = 4.8 \pm 0.39$ μM, entry 4) were approximately six-fold less potent than *N*-sulfated tobramycin **4**. Surprisingly, both heptasaccharide **9** ($IC_{50} = 41 \pm 3.6$ μM, entry 5) and octasaccharide **10** ($IC_{50} = 3.9 \pm 0.20$ μM, entry 6) bearing the GlcN-GlcA and GlcN-IdoA disaccharide cores exhibited significantly decreased binding to heparanase compared to **4** (Table 1). Although **10** is more potent than **9**, its potency is approximately five-fold lower than that of **4** and similar to both *N*-sulfated kanamycin **6** and apramycin **8**. The observation that **4** is more potent than larger oligosaccharides **9** and **10** indicates that saccharide backbone composition, precise positioning of the sulfate groups, and overall charge play critical roles in determining the affinity of HS mimetics for heparanase.

Since *N*-sulfated tobramycin **4** exhibited the highest inhibitory activity against heparanase among the aminoglycosides tested, we chemically modified tobramycin to generate a more extensive library of potential HS mimetics with the aid of computational docking (Figure S4). Syntheses of these compounds require orthogonal protecting groups that can be chemoselectively removed to unmask hydroxyl or amino groups for selective sulfation or functionalization. This can be expedited by divergent modification of a protected pseudo-trisaccharide **11** (Scheme 1) derived in 3 steps from tobramycin (**3**). In this approach, the amines were first converted into the corresponding azides (N_3). This azide formation step should be handled with CAUTION as organic azides are potentially explosive. Next, the C6-hydroxyl in the aminosugar ring **A**, required for selective *O*-sulfation and/or functionalization, was protected with the trityl (Tr) group. The remaining hydroxyls were protected with the 2-naphthylmethyl (Nap) groups. The Tr group can be removed with trifluoroacetic acid (TFA) without affecting the Nap group. In turn, the Nap group can be cleaved with DDQ,[29] whose conditions do not affect N_3. Our recent finding illustrates that 6-*O*-sulfation of the GlcN units of HS trisaccharide **2** is critical for heparanase recognition (Figure 2A).[15] As such, the 6-*O*-Tr group was exchanged to a sulfate group, forming **12** (4 steps from **11**, Scheme 1). The 6-*O*-Tr group was also exchanged to acetyl and propargyl group forming **13** and **14**, respectively (5 steps from **11**). Next, *O*-sulfated tobramycin compounds **15**–**17** were prepared to determine their affinity for heparanase compared to their *N*-sulfated counterparts. Compound **15** bearing NH_2 was prepared in 3 steps from **11** while **16** and **17** bearing NHAc were synthesized in 4 and 5 steps, respectively (Scheme 1). In the synthesis of **15**–**17**, DDQ was used to selectively remove the Nap groups, as hydrogenolysis can reduce azide into an amine.

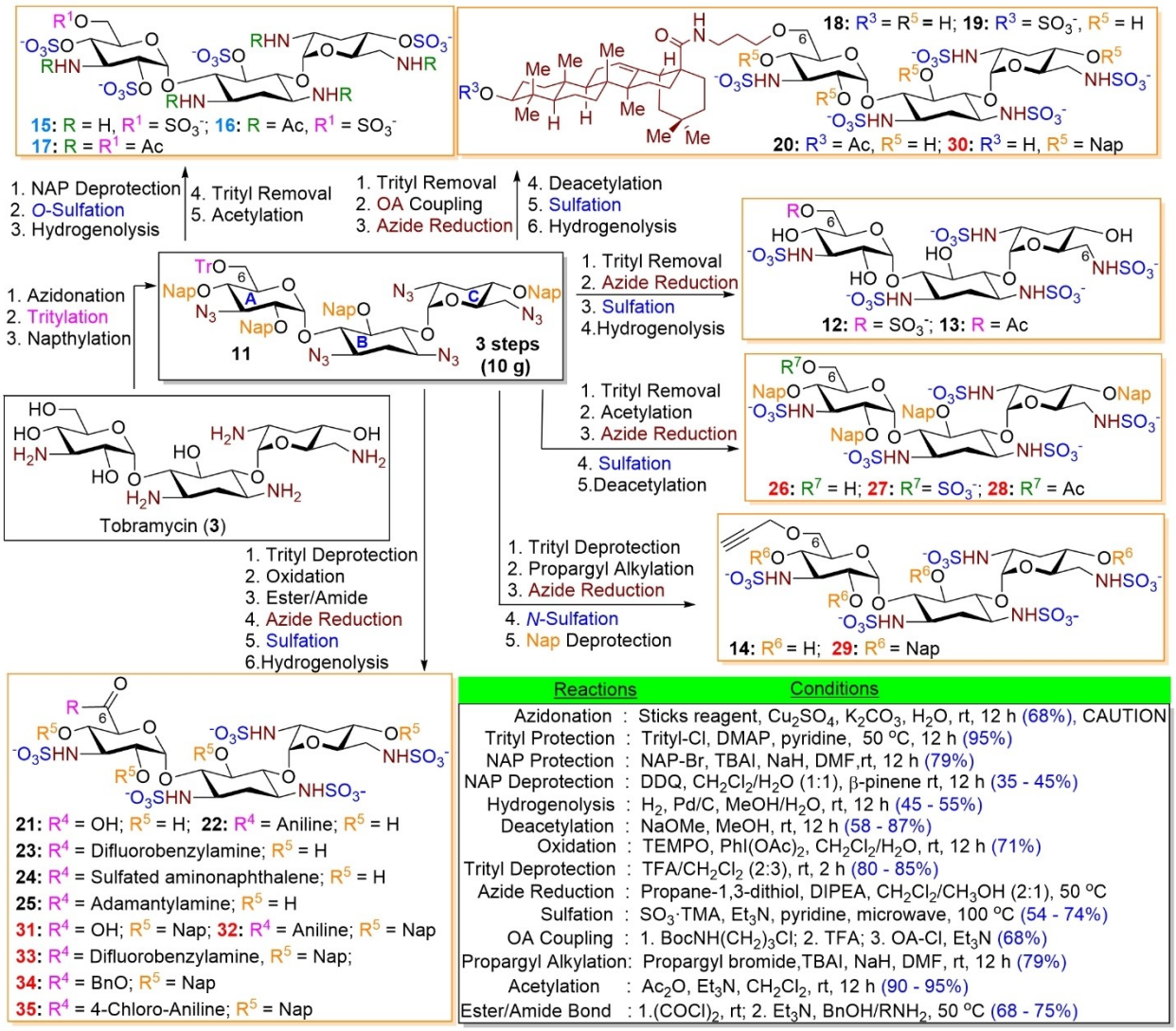
Next, we installed the lipophilic oleanolic acid linker into the C6-position of ring **A**, forming **18**–**20** (Scheme 1). Oleanolic acid has been shown to enhance anticancer activity.[30] The mechanism of action of oleanolic acid and its derivatives includes anti-cancer cell proliferation, inducing tumor cell apoptosis and autophagy, regulating cell cycle regulatory proteins, and inhibiting vascular endothelial growth, tumor cell migration, and invasion. Further, the attachment of a lipophilic linker to the sugar backbone has been reported to improve potency and pharmacokinetics and reduce common side effects of HS mimetics.[31] After removing the trityl group, the free C6-hydroxyl in ring **A** was alkylated with $BocHN(CH_2)_2Cl$, followed by Boc deprotection and amidation with an oleanolic acid chloride. This trisaccharide-oleanolic acid intermediate requires only four sequential steps (azide reduction, sulfation, hydrogenolysis, and deacetylation) to generate **19**. It takes 3 steps from the trisaccharide-oleanolic acid intermediate to form **20** (azide reduction, sulfation, and hydrogenolysis). Subsequent deacetylation of **20** provides compound **18**.

Table 1: Inhibition of heparanase by *N*-sulfated pseudo-trisaccharide aminoglycosides **4**, **6**, and **8**, HS trisaccharide **2**, and large oligosaccharides **9** and **10**.

Entry	Sulfated oligosaccharides	IC_{50} values
1	HS trisaccharide **2**	0.39 ± 0.010 μM
2	*N*-sulfated tobramycin **4**	0.88 ± 0.030 μM
3	*N*-sulfated kanamycin **6**	5.9 ± 0.77 μM
4	*N*-sulfated apramycin **8**	4.8 ± 0.39 μM
5	GlcNS6S-GlcA-[GlcNS6S-IdoA2S]$_2$-GlcNS6S-GlcA—O—  —NO_2 **9**	41 ± 3.6 μM
6	[GlcA-GlcNS6S]$_2$-IdoA2S-GlcNS6S-GlcA—O—  —N(H)—C(=O)—$(CH_2)_5$—N_3 **10**	3.9 ± 0.20 μM

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Scheme 1. Synthesis of sulfated trisaccharides from tobramycin.

The crystal structure of heparanase-HS ligand[25] and our recent docking study of HS trisaccharide **2** (Figure 4A) indicate that the CO_2^- group of the GlcA is vital for recognition due to its hydrogen bonding with Lys231 and Lys232 (Figure 4A). As a consequence of these findings, the C6-hydroxyl in ring **A** was converted into carboxylic acid **21** (Scheme 1). This was followed by the amidation of **21** with several anilines and aliphatic amines to create compounds **22**–**25**. It takes 5 steps to prepare **21** from the key intermediate **11** and 6 steps to synthesize **22**–**25** (Scheme 1). Heparanase possesses a single deep binding groove.[25] Thus, its binding pocket could accommodate the aromatic 2-naphthylmethyl (Nap) protected compounds. We hypothesize that the Nap group can stabilize the ligands through hydrophobic and cation-pi interactions with amino acid residues in the binding pocket of heparanase (Figure S4). The divergent transformations of the intermediate **11** to ten compounds (**26**–**35**) were achieved in 3–6 steps.

Overall, this approach expedites the synthesis of 27 HS mimetics with defined *O*- and *N*-sulfations in 7–12 steps using a single strategically key-protected trisaccharide **11** (10 g scale) derived from natural tobramycin. All sulfated compounds were purified by Sephadex chromatography, confirmed by NMR and MS, and produced on a 10–100 mg scale. Their ability to inhibit heparanase was assessed (see Scheme S4 for IC_{50} values of all compounds) using the same TR-FRET assay for evaluating **4**, **6**, and **8**–**10** (Table 1). We first assessed the significance of *N*-sulfated compounds bearing the 6-*O*-sulfate (**12**), acetyl (**13**), and propargyl (**14**) on ring **A** as well as *O*-sulfated compounds **15**–**17**. Unlike HS trisaccharide **2** whose 6-*O*-SO_3^- is critical for binding to heparanase,[15] the presence of the 6-*O*-SO_3^- on ring **A** of **12** has negative impacts. Compound **12** ($IC_{50} = 4.0 \pm 0.55$ μM, Figure 5) was much less potent than *N*-sulfated tobramycin **4** ($IC_{50} = 0.88 \pm 0.030$ μM) and HS trisaccharide **2** ($IC_{50} = 0.39 \pm 0.010$ μM) against heparanase. This result illustrates that the number of sulfates does not correlate with

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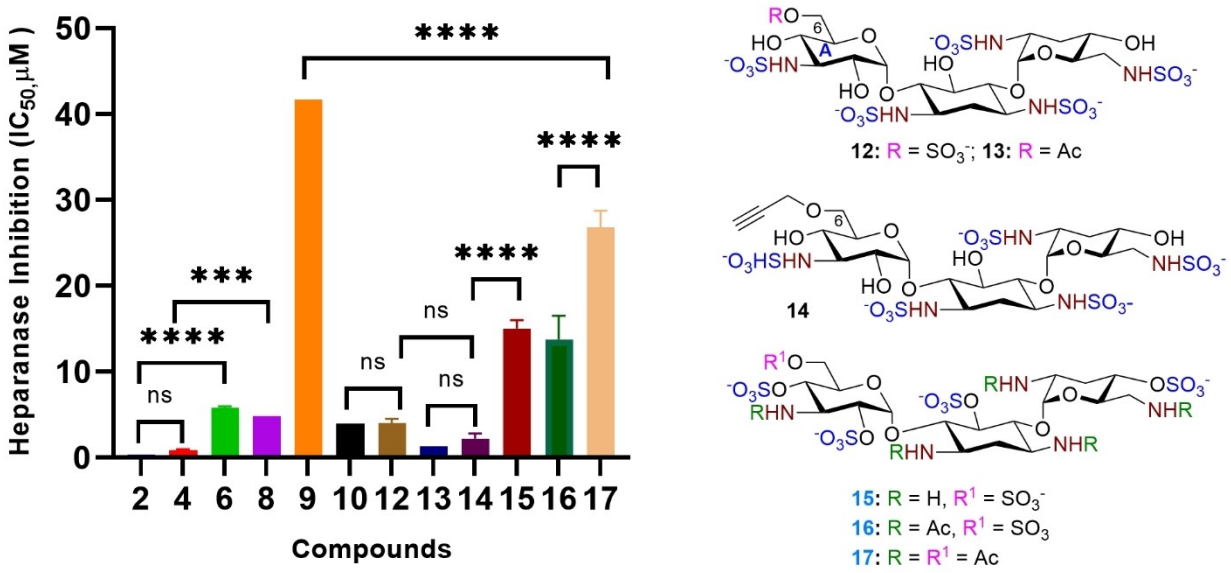


Figure 5. Inhibition of heparanase activity by sulfated tobramycin derivatives using a TR-FRET assay. Data are shown as means ± SEM ($n = 3$). *** $p < 0.0005$ and **** $p < 0.0001$.

heparanase binding. Docking analysis showed that **12** (Figure S9) is oriented in the opposite direction of **2** and **4** in the binding pocket of heparanase to minimize steric interactions of the 6-*O*-SO_3^- on ring **A** with amino acid residues lining the binding pocket. A similar effect was observed with **13** ($IC_{50} = 1.3 \pm 0.11$ μM) and **14** ($IC_{50} = 2.2 \pm 0.33$ μM). Nevertheless, **12**–**14** were more potent than heptasaccharide **9** and displayed similar inhibitory activity to octasaccharide **10** (Table 1). Next, we compared the impacts of *O*-sulfated compounds **15**–**17** to *N*-sulfated compounds (Figure 5). *O*-Sulfated compound **15** displayed modest potency ($IC_{50} = 15 \pm 1.2$ μM, Figure 5) compared to *N*-sulfated compound **4**. While **4** is oriented in the binding pocket of heparanase analogous to HS trisaccharide **2**, compound **15** was located in the opposite direction (Figure S4). Other *O*-sulfated compounds **16** ($IC_{50} = 14 \pm 0.99$ μM) and **17** ($IC_{50} = 27 \pm 0.98$ μM) also exhibited modest inhibition. These data indicate that *O*-sulfated compounds are less potent heparanase inhibitors than *N*-sulfated compounds.

Next, we evaluated **21**–**25** for their inhibitory activity against heparanase (Figure 6A). Compound **21** bearing the C6-CO_2^- group was the most potent heparanase inhibitor with $IC_{50} = 50 \pm 1.2$ nM. The presence of the CO_2^- group at the C6 position on ring **A** significantly impacts the binding affinity, as evident from the comparison of **21** with *N*-sulfated compound **4** bearing the C6-hydroxyl group (Table 1). Although the presence of the C6-aniline amide group in **22** increases heparanase binding compared to **4**, compound **22** ($IC_{50} = 180 \pm 0.75$ nM) is less potent than **21**. In contrast, the presence of the C6- difluoro-benzyl amide in **23** ($IC_{50} = 17 \pm 1.2$ μM), sulfated aromatic amide in **24** ($IC_{50} = 88 \pm 1.9$ μM), and adamantyl amide in **25** ($IC_{50} = 8.5 \pm 0.93$ μM), reduces the potency to micromolar concentrations compared to **21** and **22**.

Computational studies of compounds **21** and **22** were performed. At the end of MD, **21** has strong interactions with numerous residues deep in the binding pocket (Figure 6B). This includes 21 specific types of hydrogen bonding throughout the binding pocket with a high percentage of occupancy (>20 %) over the course of the simulation (Table S1) compared to all other compounds, which show 0–15 hydrogen bonds <20 % (Tables S2–S7). Due to its small size, **21** shows unique interactions with Asn224 and Glu343 deep inside the pocket (Figure 6B) that help to explain its high affinity. Compound **22** has weaker average hydrogen bonding interactions over time compared to that of **21** (lowered by 10–30 % of simulation time) (Table S3). However, **22** maintains strong interactions and salt bridges with residues close to the sulfate groups (Figure 6C). The presence of the C6-phenyl group pushes **22** to charged interactions on the surface of the binding pocket (Figure 6C). In addition, Arg303 curves over the top of the binding pocket as part of its interaction, closing off that section and potentially disfavoring unbinding. Among compounds screened, **24** exhibited the least potency because it made no contact with the C-terminus of heparanase (Figure S4). In contrast, the other compounds in this series showed at least two interactions with the C-terminus.

Next, we investigated heparanase binding by compounds with the lipophilic oleanolic acid at the 6-*O* position on ring **A** (Figure 7A). Although **18**–**20** have identical structures except for the functionality at the C22-position on the oleanolic acid (**18**: R = H, **19**: R = SO_3^-, and **20**: R = Ac), **18** displayed the highest inhibitory activity ($IC_{50} = 0.50 \pm 0.030$ μM) when compared to **19** ($IC_{50} = 1.3 \pm 0.10$ μM) and **20** ($IC_{50} = 1.4 \pm 0.12$ μM). Computational analysis showed that the oleanolic acid moiety of **18** is fitted into the lipophilic region of heparanase (Figure S4 in the Supporting Information). More significantly, installing the aromatic 2-

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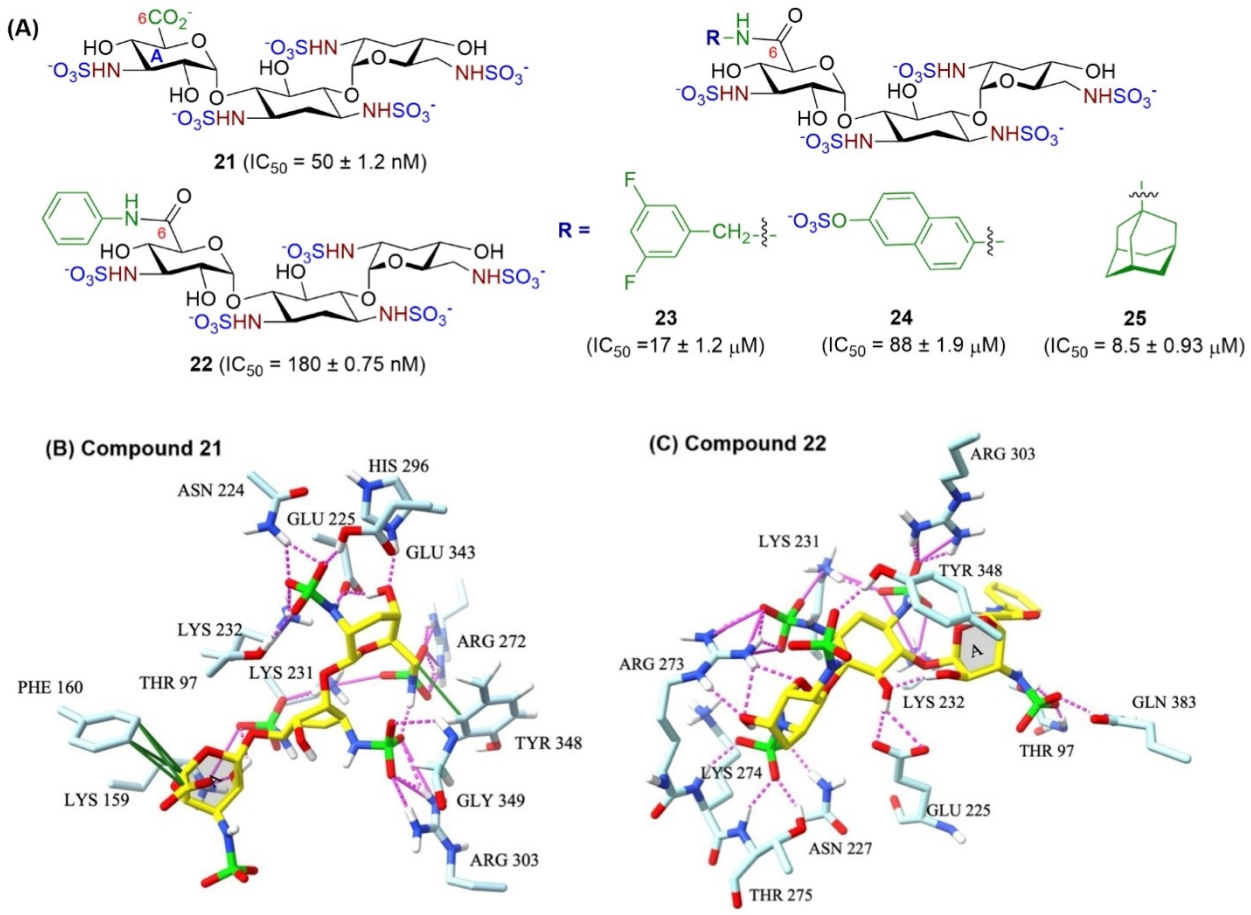


Figure 6. A) Inhibition of heparanase activity (IC_{50}) by **21**–**25** using a TR-FRET assay. B) Interactions between hydrogen bonds between **21** and key active site residues. C) Interactions between **22** for a representative snapshot at the end of the MD simulation and key active site residues.

naphthylmethyl (Nap) groups forming compound **30** proved advantageous. Compound **30** was the most potent heparanase inhibitor with an IC_{50} value in nanomolar concentration (170±2.2 nM) among compounds screened in this series (Figure 7A). Although **30** has a higher molecular weight and a more extensive ring system than **18**, the MD simulation results show that the charged section of **30** is stable and forms strong hydrogen bonds and salt bridges inside the binding pocket (Figure 7B and Table S2). The hydrophobic ring systems prevent **30** from going deep into the binding pocket like **21** (Figure 6B), but it is somewhat counteracted by the formation of pi-cation interactions between Arg272/Lys231 and the Nap rings. Additional hydrophobic interactions further stabilize **30** in the binding pocket (Figure 7B). The oleanolic acid group interacts well with His486 and Leu354 but is hampered by being primarily solvent exposed despite being predominantly hydrophobic. We also note that a well-ordered network of waters stretches across the charged region of **30** but is disrupted close to the Nap rings and oleanolic acid moiety. This could indicate that part of the reason for **30**'s high affinity involves the reduction of desolvation energy required for **30** to bind to the surface of the protein.

Most of the 2-naphthylmethyl (Nap)-containing compounds show good affinity to heparanase (Figure 8A). In addition to the oleanolic acid-containing compound **30** (IC_{50} = 170±2.2 nM), compounds **29** (IC_{50} = 190±2.3 nM) and **34** (IC_{50} = 120±0.91 nM) are among the most potent inhibitors of heparanase. Unlike **30** (Figure 7B), compound **34** has lost some hydrogen bond interactions at the substituted position with the new benzyl ester group (Figure 8B). However, **34** has gained several new strong hydrogen bond interactions with Ser228 and Thr447 (Table S7). In addition, unlike **30**, compound **34** maintains strong interactions deep inside the pocket, which helps to ensure its stability. Furthermore, **34** has gained strong pi-cation interactions, which pull **34** into the pocket and away from the surface solvent environment. In addition, two of the Nap ring groups and the C6 ester benzene ring stay close to Ala63, stacking with a nonpolar area of the protein and themselves, thus exposing a small nonpolar region to the solvent. This disfavors unbinding and lowers the ligand's motion to move freely in the middle region of the pocket. Compared to **21** (Figure 6B), due to the presence of the Nap groups, **34** has lost key hydrogen bond interactions (Figure S12).

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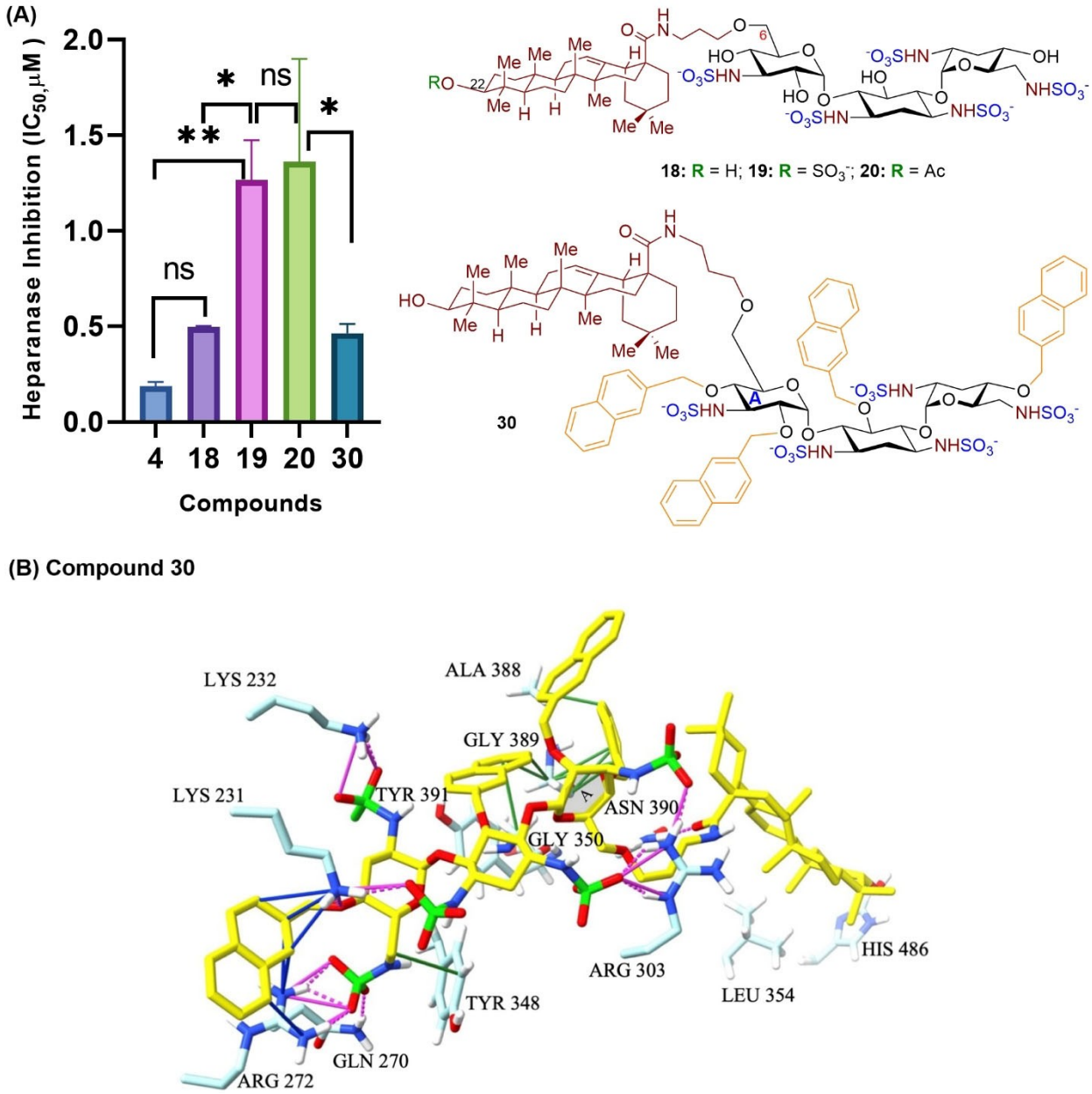


Figure 7. A) Inhibition of heparanase activity by **18–20** and **30** using a TR-FRET assay. Data are shown as means ± SEM ($n = 3$). * $p < 0.05$ and ** $p < 0.01$. B) Interactions between compound **30** at the end of the MD simulation.

Compound **29** maintains strong pi-cation interactions with the binding pocket of heparanase through three naphthalene groups attached to its ring A, B and C (Figure 8C). Hydrogen bonding occupancy is highest with Gln 270 and Arg 272, which makes compound **29** more stable in the binding position. Also, the high bonding occupancy accumulated near ring B makes the binding pose different to compound **21** but more like compounds **30** (Figure 7B) and **34** (Figure 8B). However, the presence of strong pi-cation in combination with hydrophobic interactions introduced with the substituted naphthalene groups make **29** more stable in the binding pocket. The presence of the Nap rings on **31** ($IC_{50} = 1.3 \pm 0.45$ μM, Figure 8A) makes it significantly less potent than **21** ($IC_{50} = 50 \pm 1.2$ nM, Figure 6A) without the Nap rings. Compound **31** has few persistent hydrogen bonds with the binding pocket (Figure S4). Its only strong interactions are pi-cation interactions at the binding pocket's edges.

Collectively, the data illustrate that compound **21**, bearing the C6-carboxylate on ring **A**, is the most potent inhibitor of heparanase ($IC_{50} = 50$ nM, Figure 6) among the 27 compounds screened. Compound **34** bearing the C6-benzyl ester on ring **A** ($IC_{50} = 120$ nM, Figure 8) is the second most potent heparanase inhibitor. Compound **30** bearing the C6-oleanolic acid on ring **A** and aromatic Nap groups ($IC_{50} = 170$ nM, Figure 7), **22** bearing the C6-aniline

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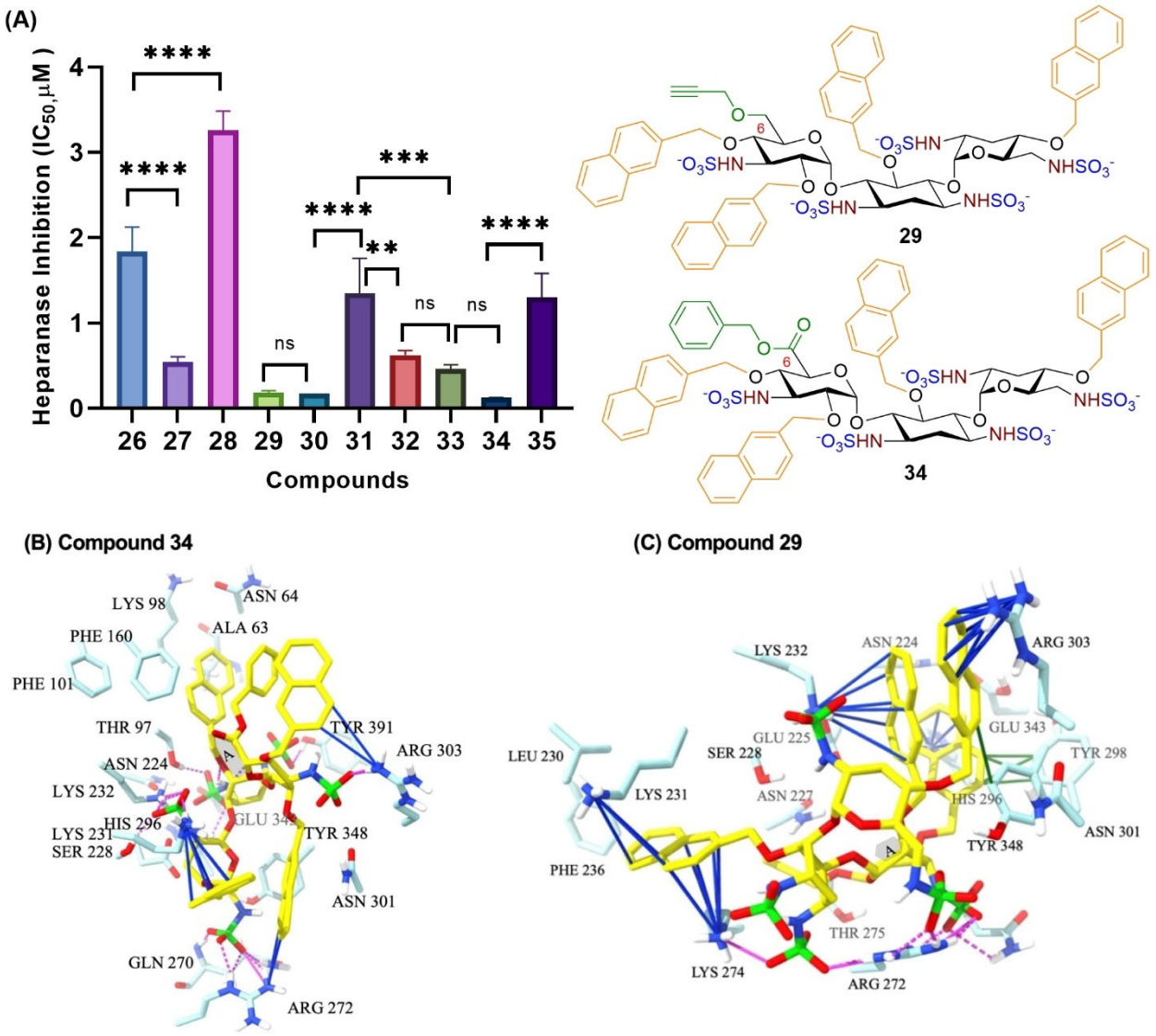


Figure 8. A) Inhibition of heparanase activity by **26–35** using a TR-FRET assay. Data are shown as means ± SEM ($n = 3$). ** $p < 0.01$, *** $p < 0.0005$ and **** $p < 0.0001$. B) Interactions between **34** for a representative snapshot at the end of the MD simulation and key residues in the binding pocket. C) Interactions between **29** for a representative snapshot at the end of the MD simulation and key residues in the binding pocket.

amide on ring **A** ($IC_{50} = 180$ nM, Figure 6), and **29** bearing the C6-propargyl group on ring **A** and aromatic Nap groups ($IC_{50} = 190$ nM, Figure 8) are the next group of potent heparanase inhibitors.

One of the critical challenges in developing HS-based therapy is minimizing the off-target activity of HS mimetics with platelet factor-4 (PF4), a naturally occurring chemokine.[32] PF4 forms a complex with heparin and HS. Heparin is known to interact with PF4 to trigger an autoimmune response causing heparin-induced thrombocytopenia (HIT).[20] In Type 1 HIT, a non-immune disorder that results from the direct effect of heparin on platelet activation. In Type 2 HIT, an immune-mediated disorder occurs after five or more days.[33] HIT is a risk factor for life-threatening thrombotic complications,[20] and is the primary reason why many anti-heparanase compounds were suspended or halted at clinical trials.[34]

Accordingly, we evaluated the ability of compounds **21**, **22**, **27**, **29**, **30**, **31**, and **34** to interact with PF4 using a solution-based biolayer interferometry (BLI) competition assay (Figures 9 and S1).[8] The most potent anti-heparanase compound **21** exhibited the highest affinity to PF4 ($IC_{50} = 79 \pm 1.9$ nM). Likewise, the 2-naphthylmethyl (Nap)-containing compound **31** also exhibited a strong PF4 binding ($IC_{50} = 200 \pm 2.30$ nM). Both **21** and **31** bear the C6-negatively charged carboxylate group on ring **A**. In contrast, the absence of the C6-carboxylate group on ring **A** reduces the affinity of other compounds to micromolar concentration compared to compounds **21** and **31** (Figure 9). For instance, compound **30** containing the C6-oleanolic acid and aromatic Nap groups exhibited 4-fold lower affinity to PF4 ($IC_{50} =$

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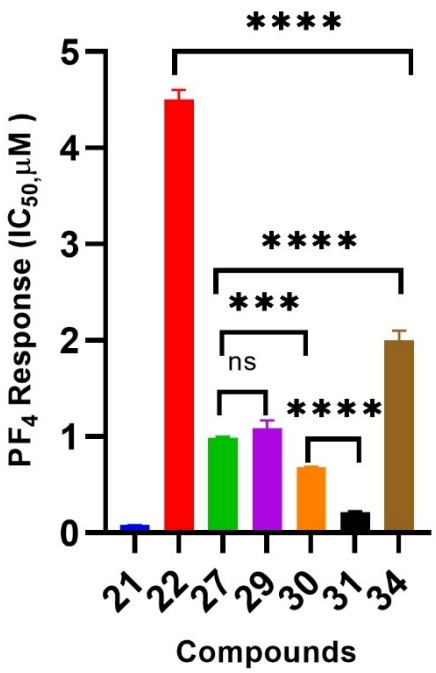


Figure 9. Interactions of PF4 with **21**, **22**, **27**, **29**, **30**, **31**, and **34** were measured using solution-based biolayer interferometry (BLI) competition assay. Data are shown as means ± SEM ($n=3$). **** $p<0.0001$.

0.68 ± 0.028 μM) than that of **31**. The second most potent anti-heparanase C6-benzyl ester **34**, binding to PF4 (IC_{50} = 2.0 ± 0.3 μM), showed nearly 25-fold lower affinity than that of **21**. On the other hand, **22** bearing the C6-aniline amide is a potent heparanase inhibitor and has the least affinity to PF4 (IC_{50} = 4.4 ± 0.36 μM, Figure 9).

Unlike heparanase, PF4 has a very surface level and solvent-exposed binding pocket, primarily stabilized by electrostatic interactions (Figure S13). Compound **22** is bound to an electronegative surface located on a side α helix. Both **21** and **34** bind to the same positions on the highly electronegative surface, while **30** is bound to a location having both electronegative and uncharged residues. Further analyzing the off-target protein, the majority of non-polar residues, including Cys (Figure S13), are located facing inside, while charged residues are located on the surface facing the solvent environment. Towards the end of the MD simulation, the hydrophobic chains of these compounds tend to move towards these nonpolar areas outside of the binding pocket.

Compound **21** forms strong hydrogen bonds and salt bridges with the binding pocket of PF4 (Figure 10A) with high hydrogen bond percent occupancy over time (> 60 %, Table S9). Compound **34** formed hydrogen bonds with His235 and Arg120 (Figure 10B), which appeared to counteract the unfavorable interaction of hydrophobic Nap rings flipping into the solvent. One Nap ring had strong hydrophobic interactions with Ile151 and was moving towards the hydrophobic area around Val119 (Figure 10B). The MD simulations of **34** appear to be highly dependent on the initial pose and relatively unstable (Figure S15). Overall, **34** is not as stable in the binding pocket as **21**, consistent with the experimental data that **34** has a lower affinity to PF4 than **21**.

Compared to **34**, compound **30** has a hydrophobic oleanolic acid substituent oriented along the trisaccharide chain relatively linearly, forming a long nonpolar "tail." Compound **30** also has a polar amide group instead of the negatively charged carboxylate group of compound **21**, which is still accessible towards the charged surface of PF4. The hydrophobic interactions made by the long hydrophobic tail allow the polar amide group to be more exposed to the protein for interactions with Pro334 (Figure 10C). Further, the Nap rings are stabilized by pi-cation interactions formed with Arg 49 and by nearby van der Waals interactions. In addition, the SO_3^- group forms salt bridge interactions with Lys331 and Lys246, making the binding stronger. Unlike **34**, compound **30** maintains these key interactions throughout the simulation. This is consistent with the experimental data that **30** has a 10-fold higher affinity to PF4 than **34**.

The binding position of **22** is different from the other ligands tested. At the end of the simulation, **22** essentially slides to the top of PF4 out of the binding pocket and towards the top cysteine residues (Figure 10D). Since this molecule moves so substantially, there is little consistency of hydrogen bond interactions between the simulation's beginning and end (Figure S17). Furthermore, the hydrophilic group exposure of **22** to the solvent in competition with the PF4 surface likely makes the binding unstable. In addition, **22** has a more nonpolar benzamide group compared to compound **21**'s carboxylate group at the same C6-position on ring **A**, making it less attracted to PF4's charged surface yet able to interact with the solvent. Unlike **34** and **30**, with a reduction of steric hindrance due to the absence of the Nap groups, compound **22** formed intramolecular hydrogen bonds, making its conformation more constrained. These data are consistent with the experimental result that **22** has the lowest affinity to PF4 among the compounds screened.

Conclusion

In summary, we have developed a systematically designed and expedient strategy to obtain a library of 27 oligosaccharides as HS mimetics from commercially available aminoglycosides. We demonstrate that the use of aminoglycoside tobramycin accelerates the synthesis of diverse collections of HS mimetics in 7–12 steps from a single trisaccharide intermediate. This approach significantly reduces the number of steps as compared to traditional strategy starting from monosaccharide building blocks. The rapid access to the 27-membered HS mimetics enabled us to probe systematically the significant impacts of saccharide structures bearing the carboxylate, amide, oleanolic acid, and the aromatic rings on selective binding to heparanase and off-target platelet factor-4. The semisynthetic approach, in combination with computational insight, enables us to identify four HS-mimicking molecules **22**, **29**, **30**, and **34** that exhibit strong binding to heparanase but have a low affinity to platelet factor-4. This expedient approach to accessing a library of

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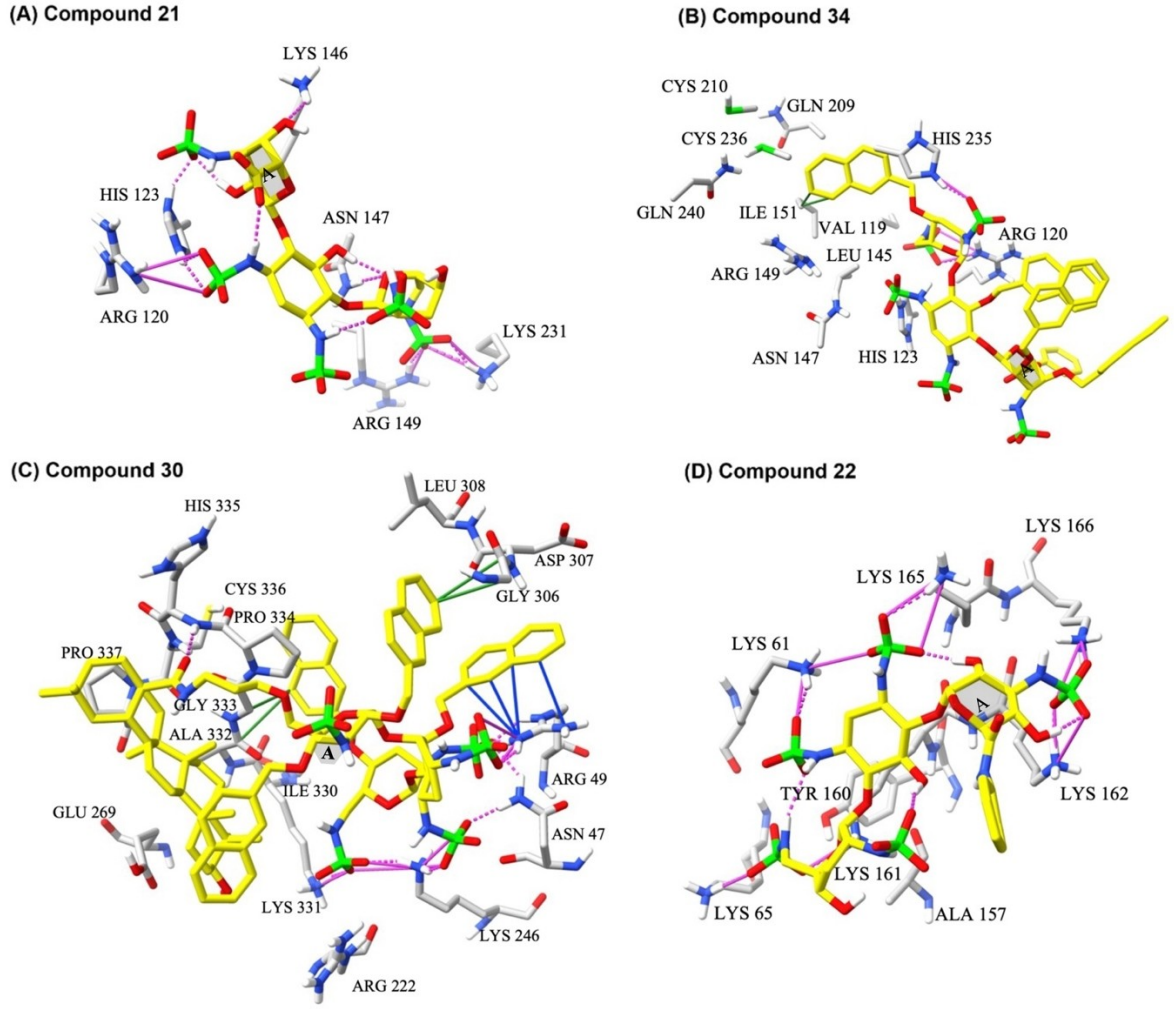


Figure 10. Interactions among compounds **21**, **34**, **30**, and **22** bound to PF4 for a representative snapshot at the end of the MD simulation and key active site residues.

HS mimetics advances our understanding of HS-heparanase and HS-PF4 interactions and provides a useful tool for developing a novel class of anti-heparanase cancer therapeutics with minimal off-target activity.

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Conflict of Interest

The authors declare no competing financial interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: Aminoglycosides · Heparan Sulfate · Heparanase · Platelet Factor 4 · Tobramycin

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